

Task 3: Linear Regression Model Development and Evaluation

Predictive Modeling — California Housing Dataset

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1. Task Overview

This task focuses on building and evaluating a supervised machine learning model for regression. A Linear Regression model is trained on the California Housing dataset to predict median house values. The task covers the full modeling pipeline: data loading, train/test splitting, model training, metric evaluation, and result visualization.

2. Dataset

The California Housing dataset is loaded directly from Scikit-learn. It contains 20,640 samples and 8 numerical features derived from the 1990 US Census, with no missing values.

Property	Value
Source	sklearn.datasets.fetch_california_housing
Rows	20,640
Features	8 numerical
Target	MedHouseVal (median house value, in \$100,000s)
Missing values	0
Train / Test	16,512 / 4,128 (80/20 split, random_state=42)

Feature	Description
MedInc	Median income in block group
HouseAge	Median house age in block group
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block group population
AveOccup	Average number of household members
Latitude	Block group latitude
Longitude	Block group longitude

Table 1: Dataset features and descriptions.

3. Methodology

Step 1 — Data Loading

The dataset is fetched using Scikit-learn's built-in loader. A DataFrame is created with named columns and the target column MedHouseVal is appended.

```
housing = fetch_california_housing()
df = pd.DataFrame(housing.data, columns=housing.feature_names)
df['MedHouseVal'] = housing.target
```

Step 2 — Train/Test Split

Data is split 80/20 with a fixed random seed (42) to ensure reproducibility.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

Step 3 — Model Training

A Linear Regression model is fitted on the training set. The model intercept is -37.0233.

```
model = LinearRegression()
model.fit(X_train, y_train)
```

Step 4 — Prediction & Metrics

Predictions are generated on the test set and three evaluation metrics are computed.

```
y_pred = model.predict(X_test)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

4. Results

Metric	Value	Interpretation
RMSE (Root Mean Sq. Err)	0.7456	Average error of ~\$74,560 per prediction
MAE (Mean Absolute Err)	0.5332	Typical prediction error of ~\$53,320
R ² (R-squared)	0.5758	Model explains 57.6% of variance in house prices

Table 2: Model evaluation metrics on the test set.

The R² score of 0.5758 indicates a moderate fit — the model captures the general trend but does not account for all sources of variance, which is expected given that Linear Regression assumes a linear relationship between features and the target. The scatter plot of predicted vs. actual values shows points clustered around the diagonal reference line, confirming a reasonable but imperfect fit. Higher-value properties tend to be underpredicted, which is a known limitation of linear models on skewed targets.

5. Visualization

A scatter plot is produced comparing actual vs. predicted house values on the test set. A red dashed diagonal line ($y = x$) represents perfect predictions — points closer to this line indicate more accurate predictions. The plot is saved as `predicted_vs_actual.png` at 300 DPI.

```
plt.scatter(y_test, y_pred, alpha=0.5, edgecolors='white', linewidth=0.5)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--',
        lw=1.5)
plt.savefig('predicted_vs_actual.png', dpi=300, bbox_inches='tight')
```

6. Deliverables

- Task3_ML_Tayyabah_Rehman.ipynb — Jupyter Notebook with all steps, comments, and outputs.
- predicted_vs_actual.png — High-resolution scatter plot of model predictions.
- README.md — GitHub documentation with setup guide, results, and interpretation.
- This PDF — Written description of the task, methodology, and results.

7. Limitations & Future Work

Linear Regression assumes a linear relationship between all features and the target, which may not hold for all features in this dataset (e.g., Latitude/Longitude have a non-linear geographic effect on house prices). Future improvements could include feature scaling before training, polynomial feature expansion, and testing non-linear models such as Random Forest or Gradient Boosting to improve the R^2 beyond 0.5758.

Dataset: California Housing (sklearn.datasets) — 20,640 samples, 8 features, no missing values.